RAK14013 WisBlock Joystick Module Datasheet

Overview

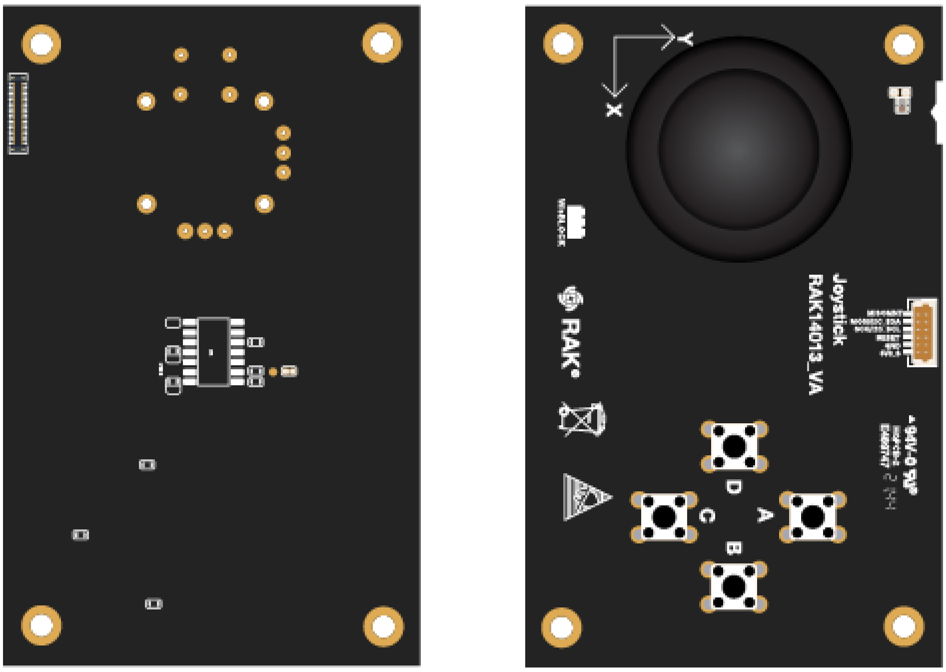


Figure 1: RAK14013 WisBlock Joystick Module

Description

The RAK14013 is a joystick module with embedded MCU based on ATTINY441-SSU from Atmel. It has four common tactile push buttons and a rotary 2-axis analog joystick. The status of the buttons and voltage readings of an analog joystick is monitored by the MCU and converted to digital data via I2C interface. RAK14013 doesn't have a WisBlock standard connector so you need a RAK14007 adapter board to plug it into to WisBlock Base module. The RAK14013 communicates to the WisBlock Core via I2C interface.

Features

- Joystick module and push buttons
- I2C interface

- 3.3 V Power supply
- Chipset: ATMEL ATTINY441-SSU
- Module Size: 54 mm x 85 mm

Specifications

Overview

Mounting

The RAK14013 WisBlock Joystick Module can be mounted to the IO slot of the WisBlock Base board. **Figure 2** shows the mounting mechanism of the RAK14006 on a WisBlock Base board, such as the RAK5005-O .

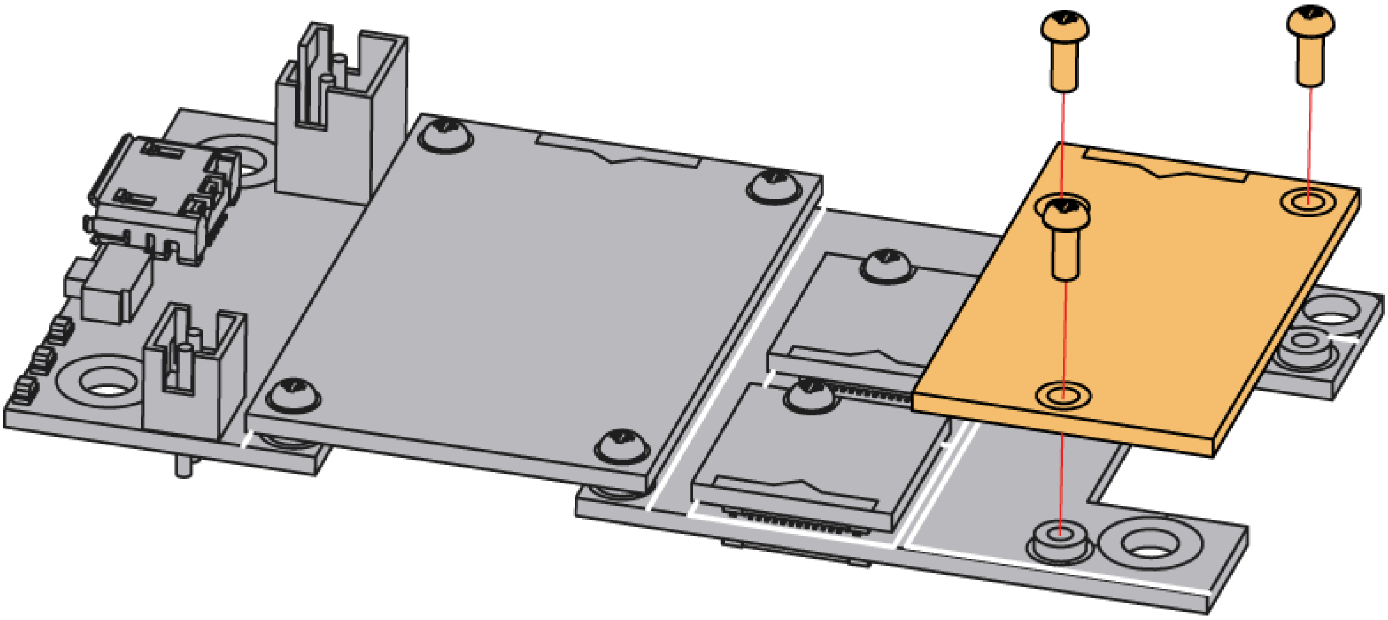


Figure 2: RAK14013 WisBlock Joystick Module mounting

Hardware

The hardware specification is categorized into three parts. It provides the pinouts of the module and their corresponding functions and diagrams. Also, it shows the mechanical parameters of the RAK14013 WisBlock Joystick Module.

Chipset

Vendor	Part number
ATMEL	ATTINY441-SSU

Pin Definition

The pin definition of the RAK14013 connector is shown in **Figure 3**.

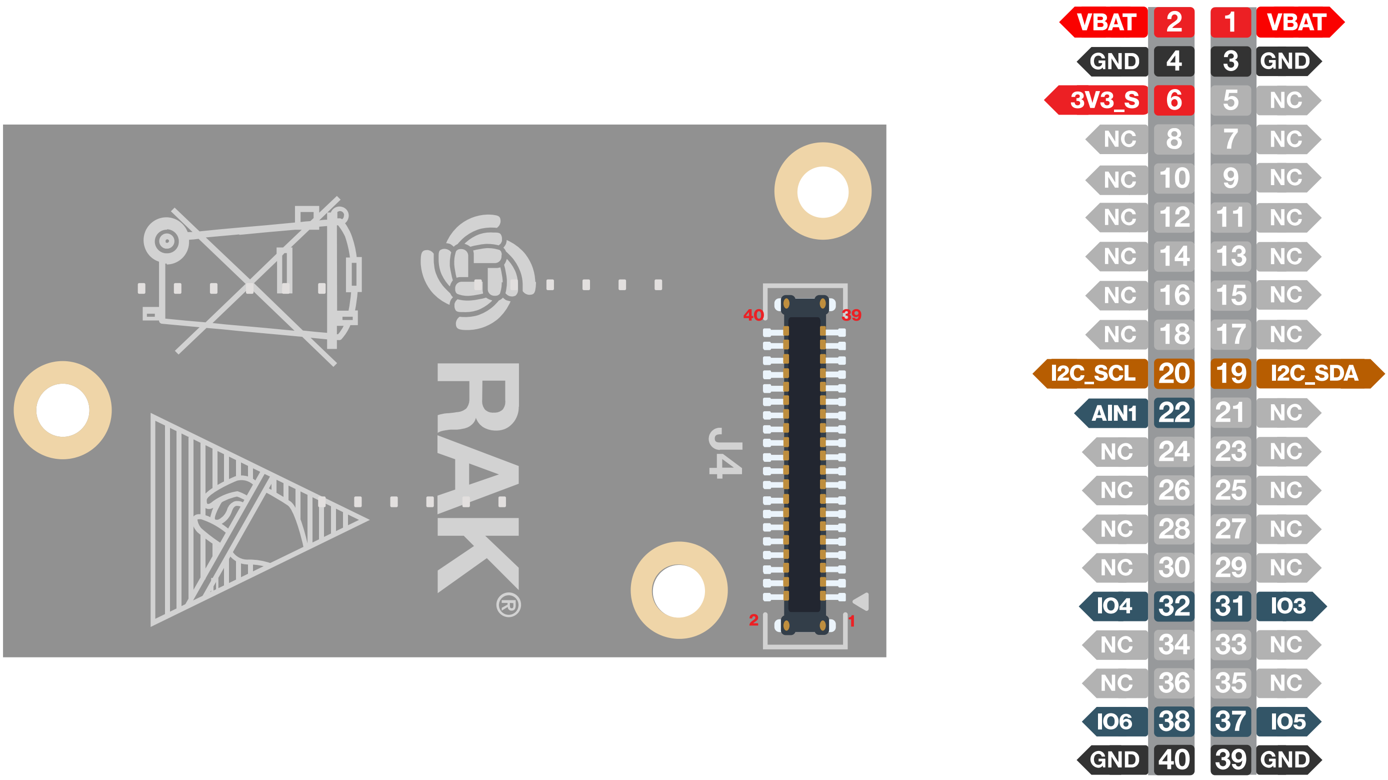


Figure 3: RAK14013 WisBlock Joystick Module pinout

> **NOTE**
>
> - **3V3**, **I2C** related pin, **RESET**, and **INT** are connected to cable
> - This connector is also for updating the MCU software

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanical drawing of the RAK14013 module.

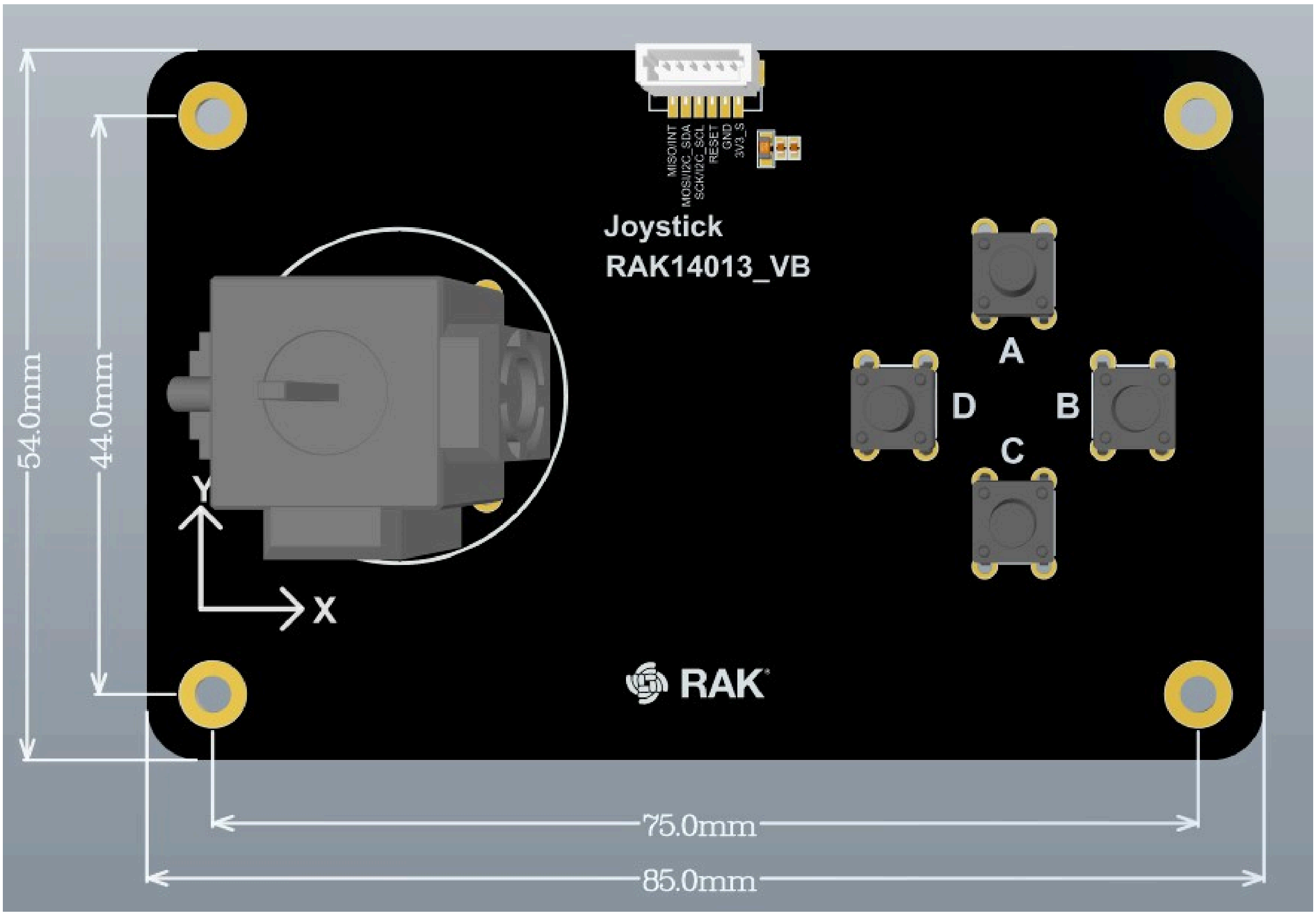
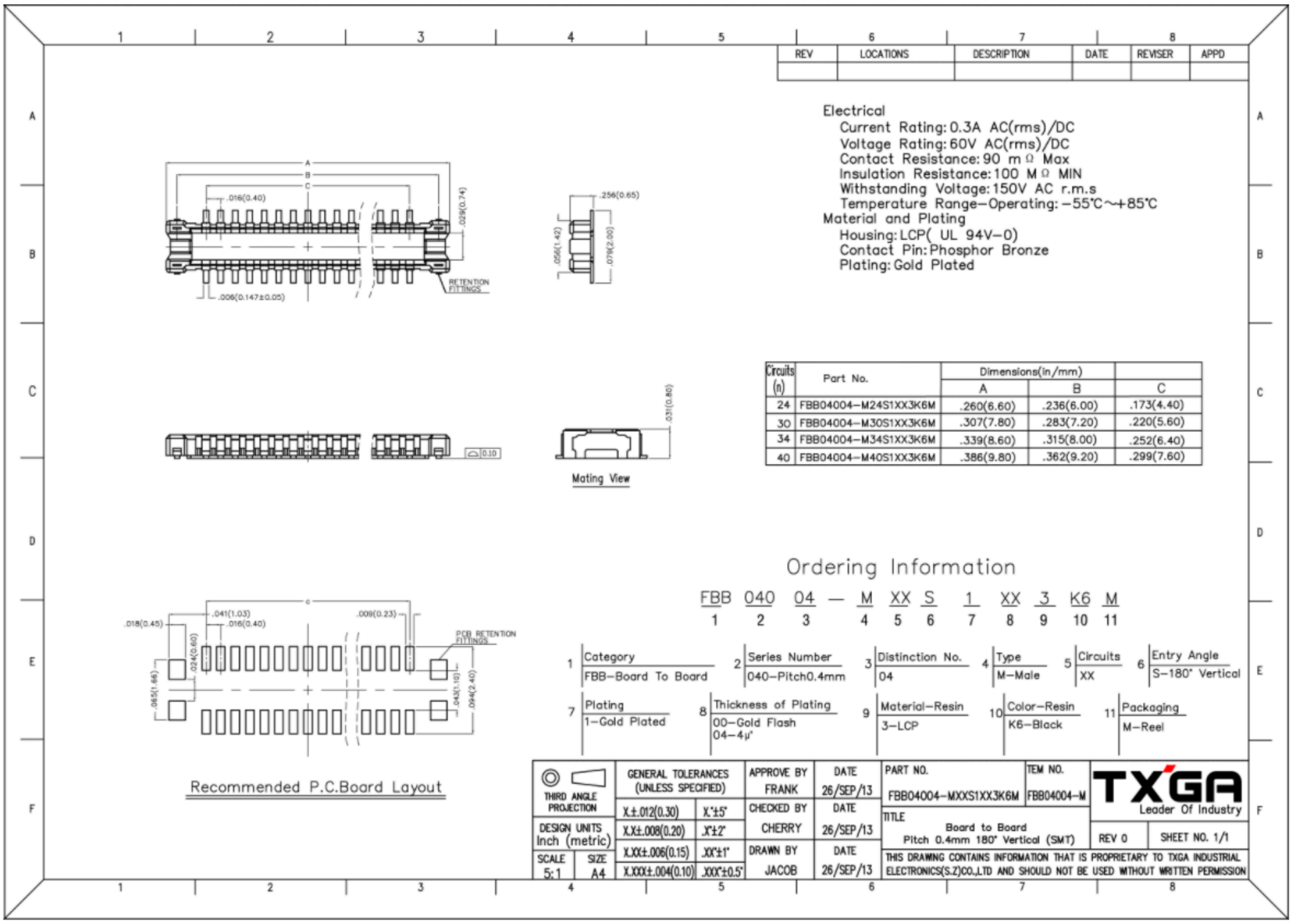


Figure 4: RAK14013 WisBlock Joystick Module Dimensions

WisConnector PCB Layout



Circuits (n)	Part No.	Dimensions(in/mm) A	B	C
24	FBB04004-M24S1XX3K6M	.260(6.60)	.236(6.00)	.173(4.40)
30	FBB04004-M30S1XX3K6M	.307(7.80)	.283(7.20)	.220(5.60)
34	FBB04004-M34S1XX3K6M	.339(8.60)	.315(8.00)	.252(6.40)
40	FBB04004-M40S1XX3K6M	.386(9.80)	.362(9.20)	.299(7.60)

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK14013 WisBlock Joystick Module.

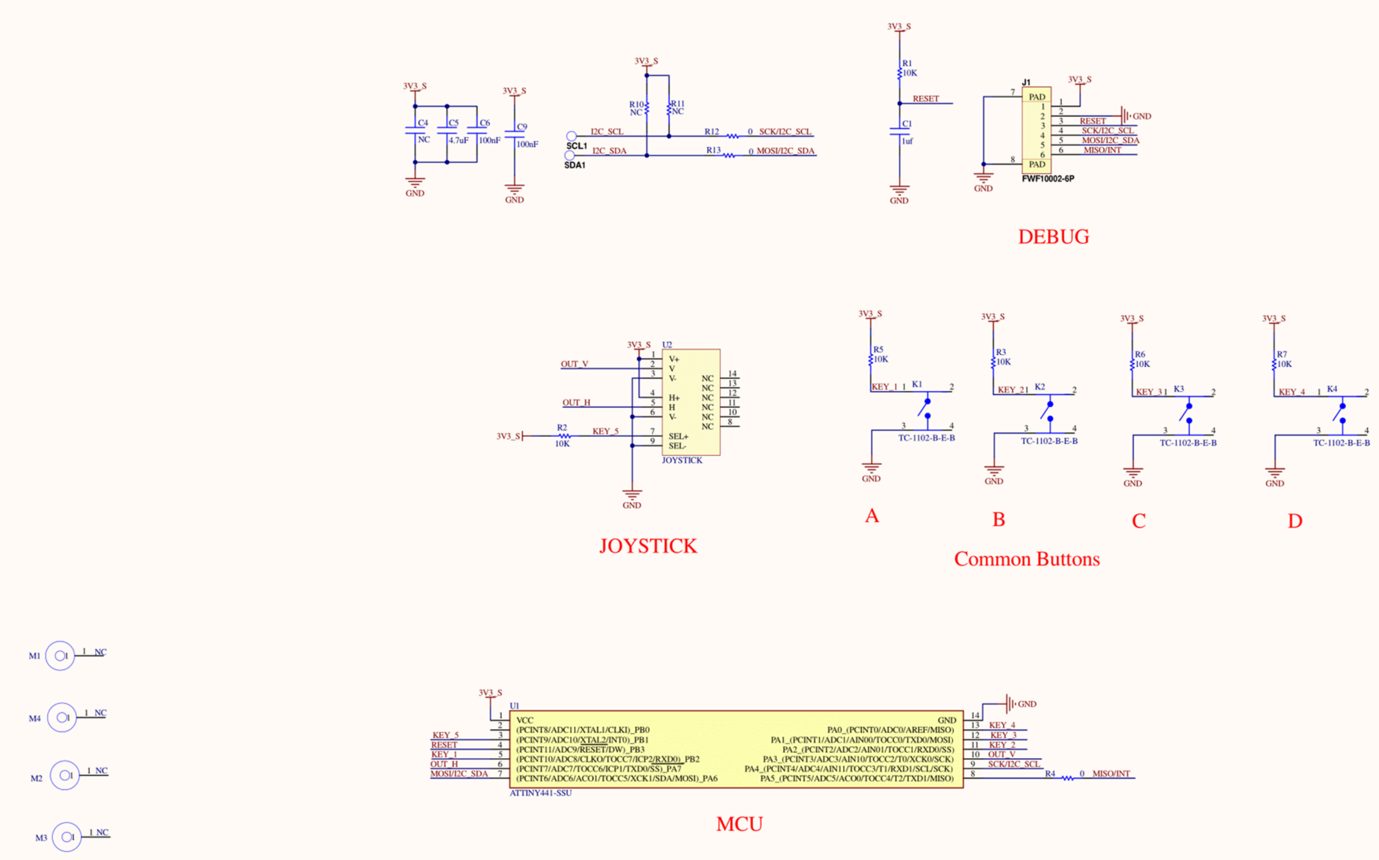


Figure 6: RAK14013 WisBlock Joystick Module schematic

Power Supply and I2C

Power supply and **I2C** connections are shown in **Figure 7**.

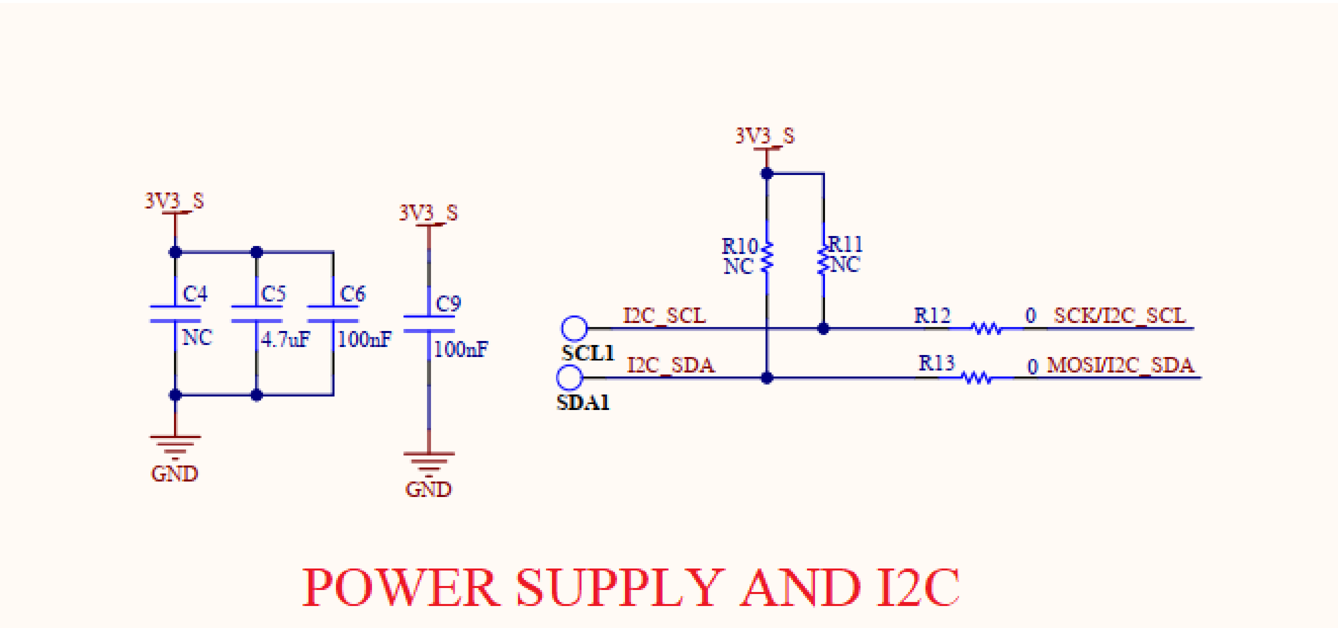


Figure 7: RAK14013 Power supply and I2C

MCU Circuit

The RAK14013 uses ATTINY441-SSU microcontroller unit (MCU) to control the four tactile push buttons and a rotary series potentiometer of the joystick.

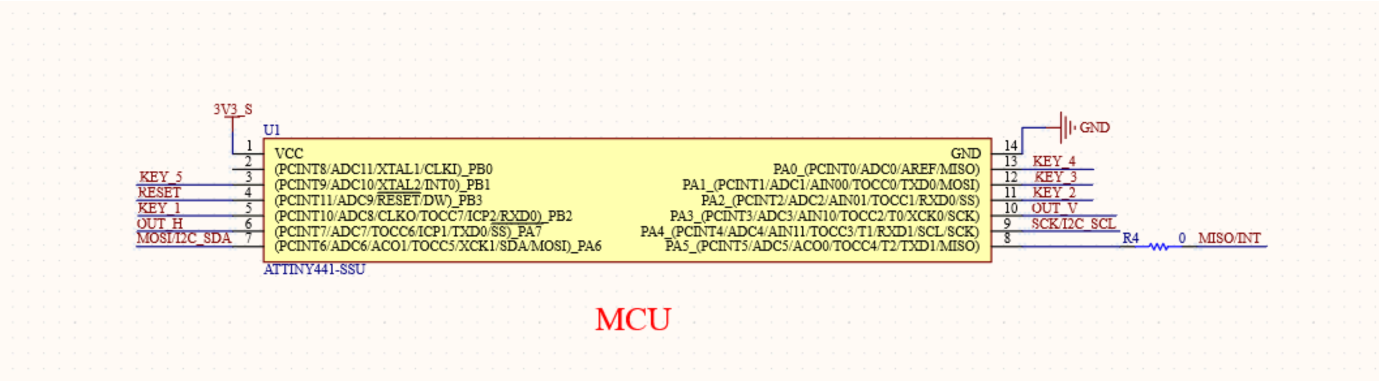


Figure 8: MCU circuit

Joystick Circuit

Joystick circuit diagram is shown in **Figure 9**.

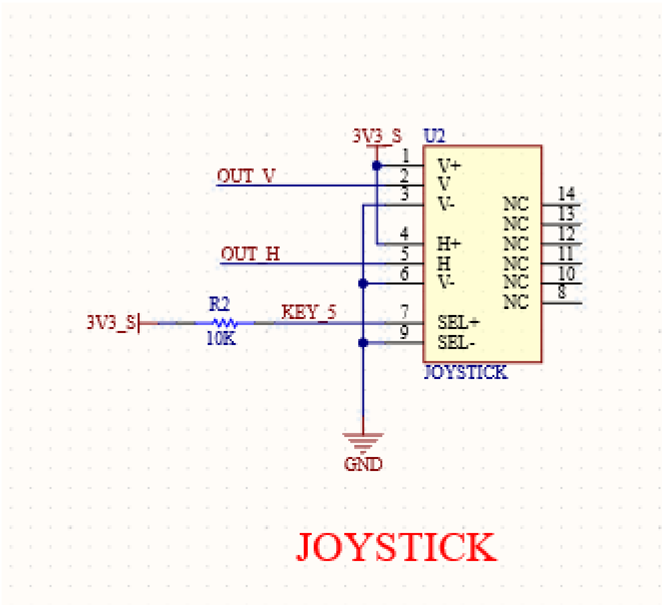


Figure 9: RAK14013 joystick

Tactile Pushbuttons Circuit

Tactile Pushbuttons circuit diagram is shown in **Figure 10**.

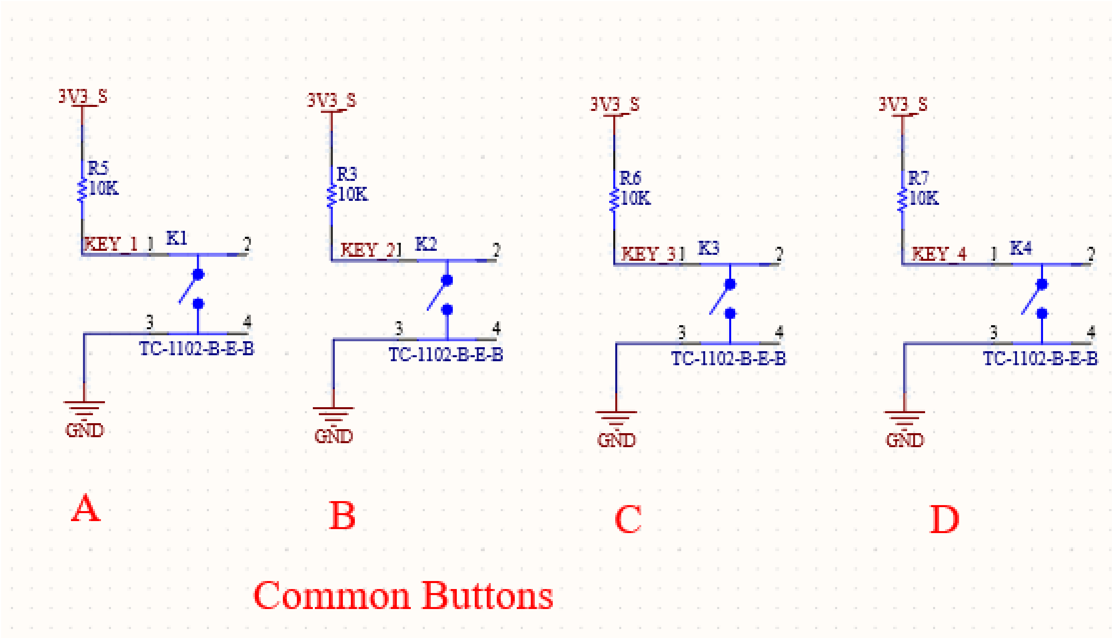


Figure 10: RAK14013 tactile push buttons

Debug Connector

Debug connector circuit diagram is shown in **Figure 11**.

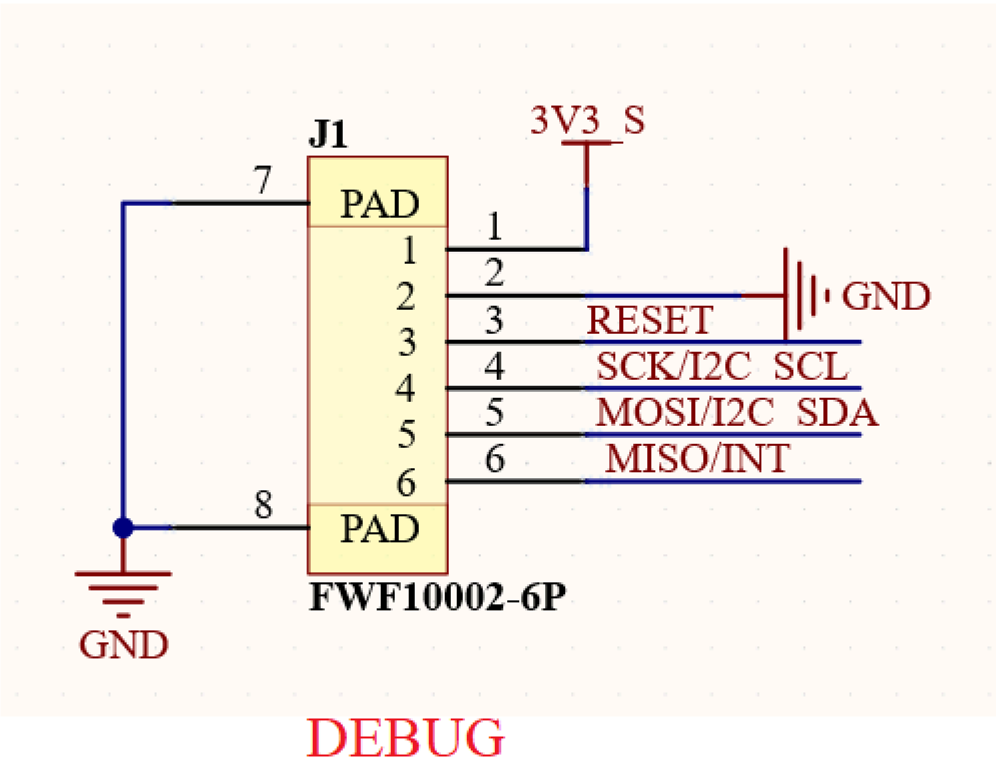


Figure 11: RAK14013 debug connector